Streamlined Performance Testing with Test Containers

Andrew Clegg

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5

Andrew Clegg

- Been at SAS since 2014
- Performance Engineer
- MISFITs team
- Two dogs
- Ultimate

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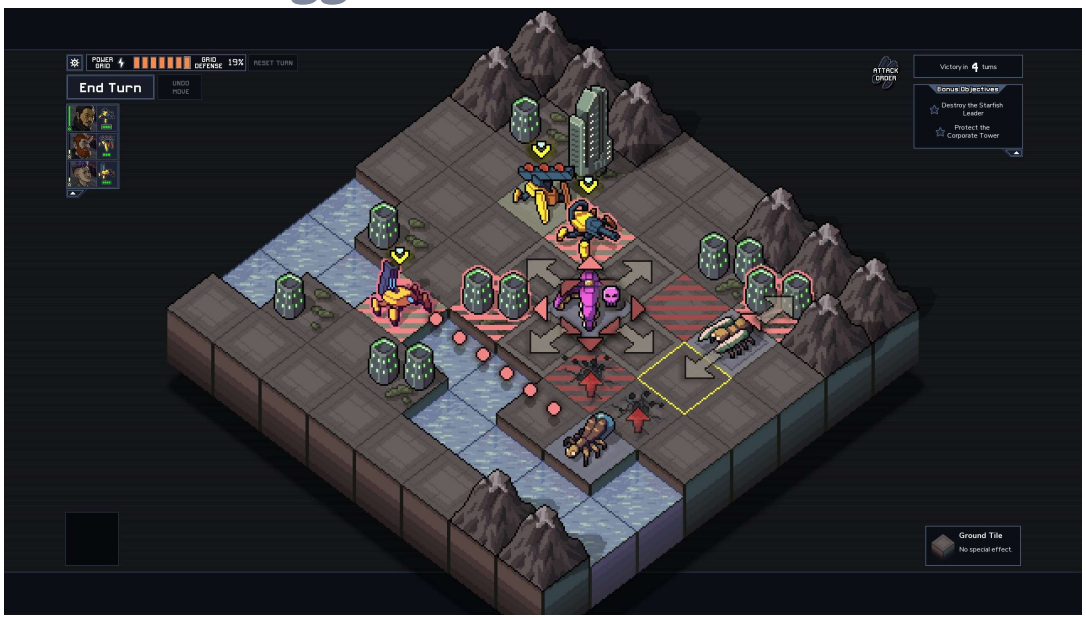

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To Performance Engineers

- How to get involved in functional tests
- Trust their expertise

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To SDETs

- How you can get involved in performance testing

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Ye Olden Performance Testing

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2014

- Graduated college
- Joined performance engineering team
- One performance test tool
- Each week we would re-record our workflow

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API recordings

- Browser recordings for API traffic
  - Careful where you click or you collect extra calls that you have to correlate
  - Ignores the client-side

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Attempts to make test recording less painful

- One of the first tools I wrote was an auto-converter for LR API scripts
  - JobIDs
  - CSRF/Session tokens
  - Breakages with every fresh deployment
  - Spend lots of time writing a few test cases
- Ranorex
  - First foray into UI testing
  - Wrote a lot of supporting software

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The problem of scale

- Limited numbers of performance engineers
- New features
- API changes
- Edge-cases
- Variety of products all developing independently
  - One perf team to handle all of these different cases
  - After getting assigned to product, learn that product and can use that test-container environment for other teams
  - Awakening

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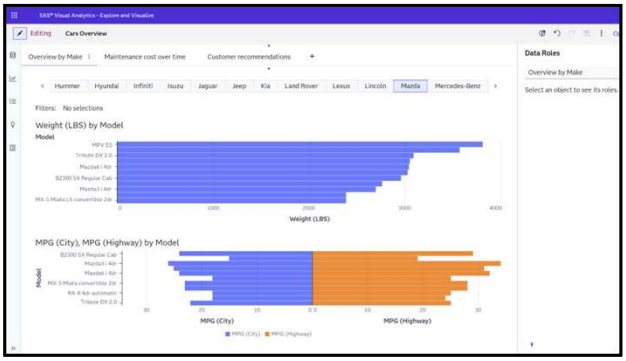
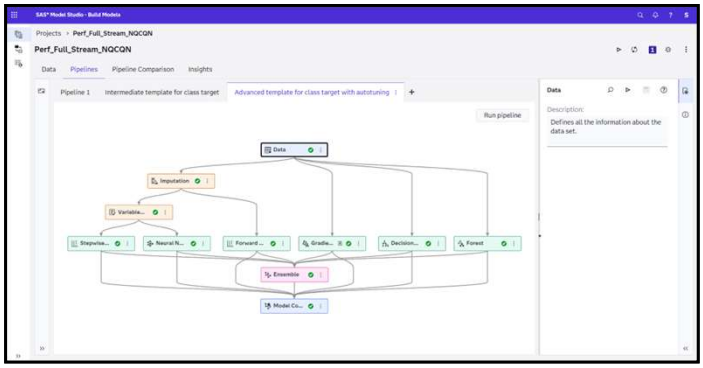
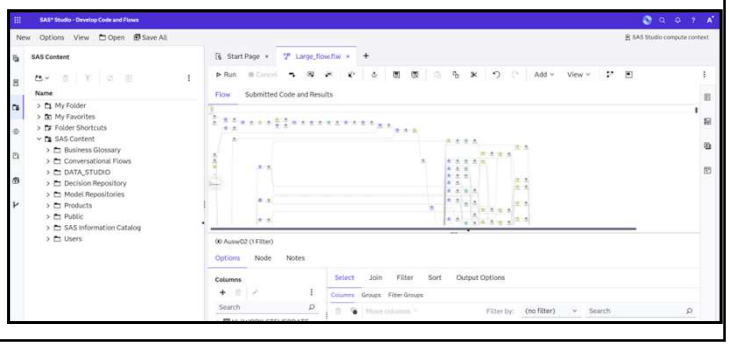
Product variety

- Show code for Studio







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Reorganization

- Team structure changes
  - Used to have dedicated performance team
    - Insular, built our own tools
    - No idea what was going on in the outside world
  - Reorg occurred, perf engineers split to the product teams

  - TO BE DELETED

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Using existing expertise

Stand on the shoulders of giants

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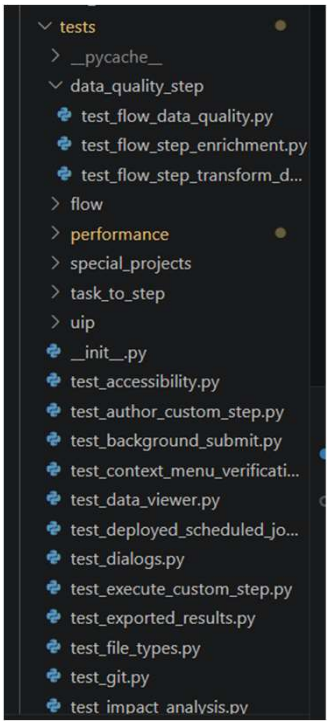
Test Containers

- Already existed with high-quality tests
- Very specific to the product
- Not generalized
  - higher value of testing a specific product/area
- Standardized base container / test container framework
  - Standard entrypoints
  - WHAT IS A TEST CONTAINER
    - Standard bits – SAS requires Lacuna as the test container base



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Reuse expertise

- Every interaction ever needed is already done
  - Or should be!
- Give examples of UI pieces –
  - Search – Why am I rewriting an xpath for the search input, search results, validating that we got 10 items back
    - This already exists in the functional tests
    - Graphic to show this
- Performance engineers get to focus on performance
  - Can request workflows if needed, and get help from SDETs
  - VP requested a specific test to hit …something, and we built it in 30 minutes

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K8s loadgen / pyperfhandler

"Is this open source?" "No"

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K8s loadgen / pyperfhandler interactions

Both internal tools

- Pyperfhandler
  - Instrument python tests in a standardized format
- K8s-lg
  - Run containers at scale
  - Organize data visualization

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Test Containers

Performance

```
Then the user selects the 'Log' tab in 'Expression Builder' dialog
Then the log tab has '0' 'errors'
Then the log tab has '0' 'warnings'
Then the log tab has '10' 'notes'
Given the 'Save baseline expression builder' transaction is started
Then the user clicks 'Save' to close the expression builder
And the 'filtered' count is: '11'
Then the 'Save baseline expression builder' transaction is ended
And the cell value 'Atlanta' exists on row '1'
Then the user clicks the toolbar 'Clear entry' button
And the cell value 'Cleveland' exists on row '1'
And the 'rows' count is: '322'
```

```
with studio_main_page.performance.transaction("Retrieve compute context"):
    # Wait for the compute context to be ready, max 6 minutes
    studio_main_page.compute_context_button.wait_for(timeout=360000)
```

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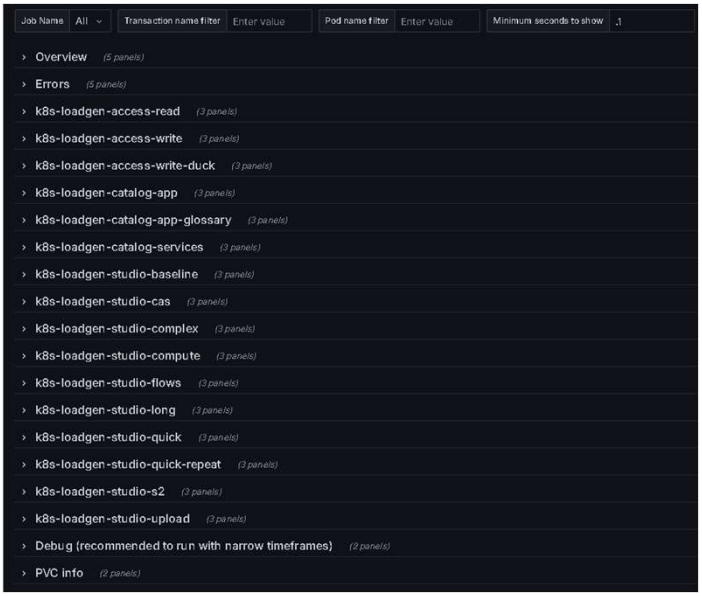
Workflow list

- Widespread, long-running integration tests
- Limited full-time monitoring requirements



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- Video recording showing k8slg in use

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Pivot

Can we just … run those existing tests at scale?

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Cool projects we've done as a result

- Widespread integration testing efforts
  - Without each individual having to manage their test results
- Chaos testing
  - Take down an AWS/AKS/GCP region and see how products handle it
  - Don't need all these functional testers to manually hop on a system, or kick off their tests at a random system each week
  - Use k8s-lg and functional tests -> point them at the cluster and get results

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Results

- Unique bugs found?

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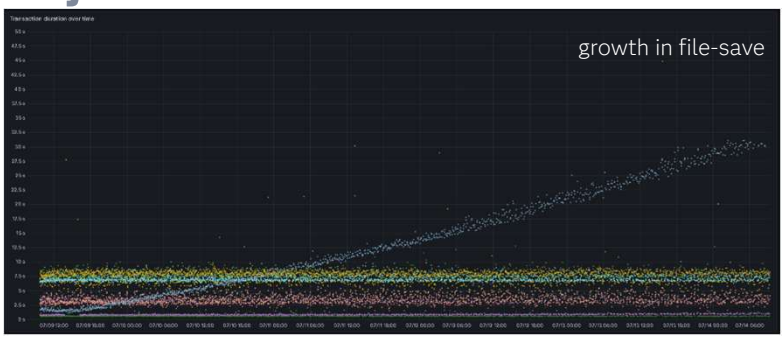
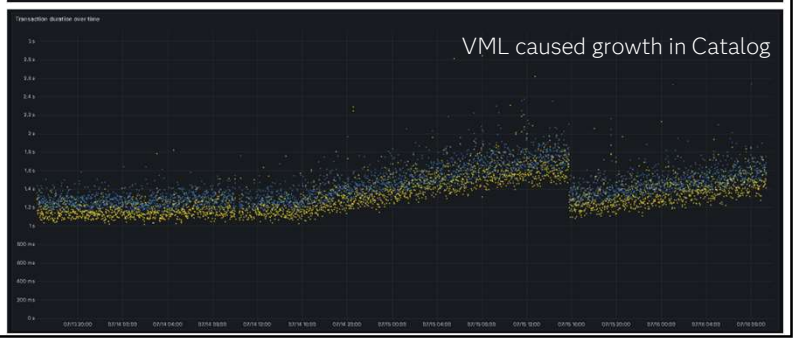
Endurance Testing Project

Stability week integration





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File browse investigation

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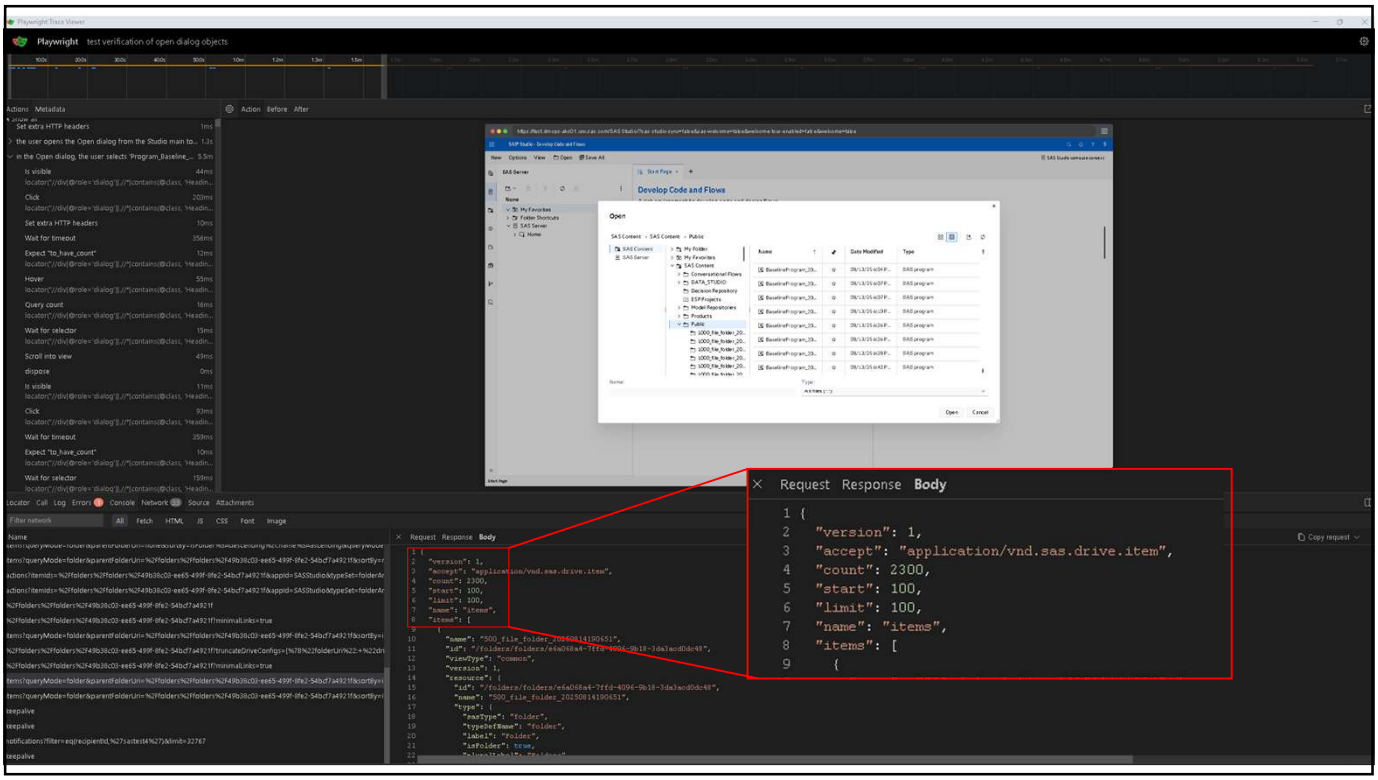

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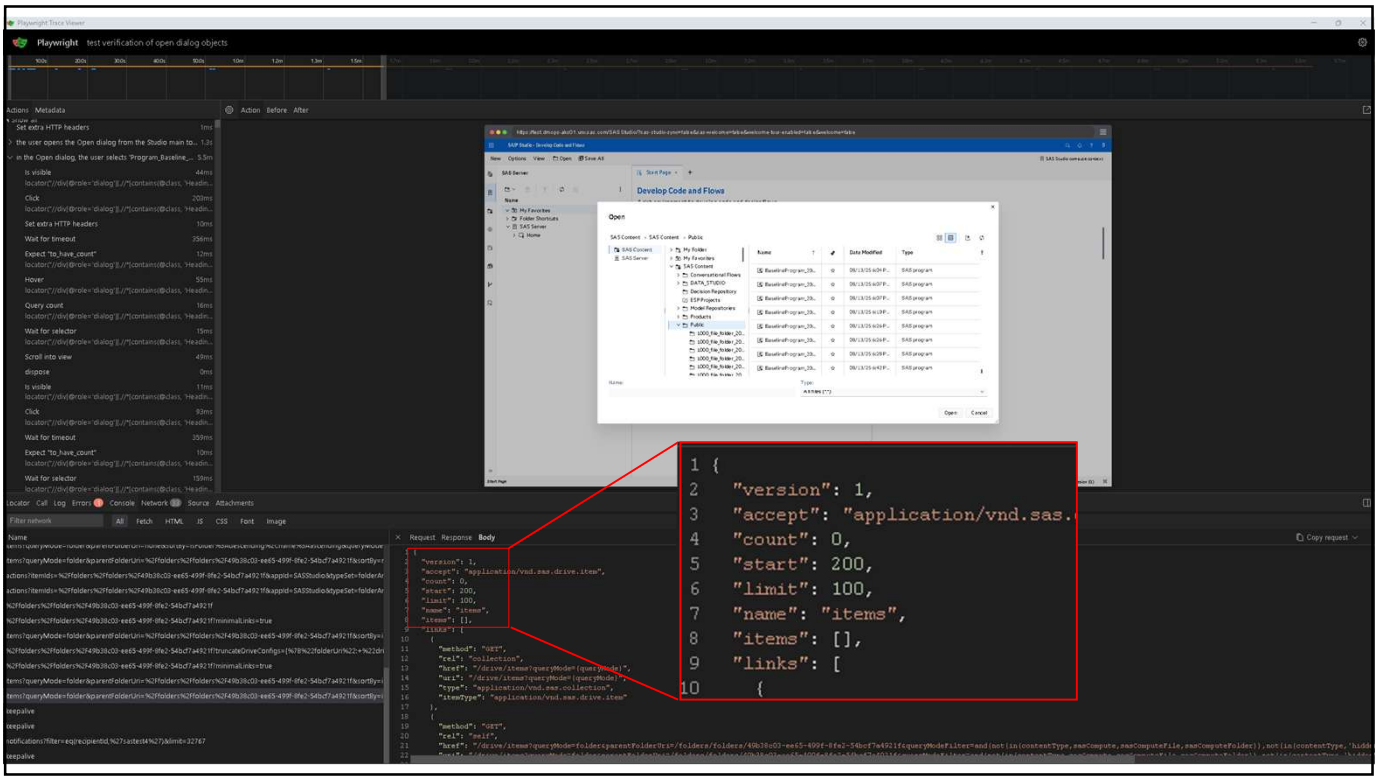

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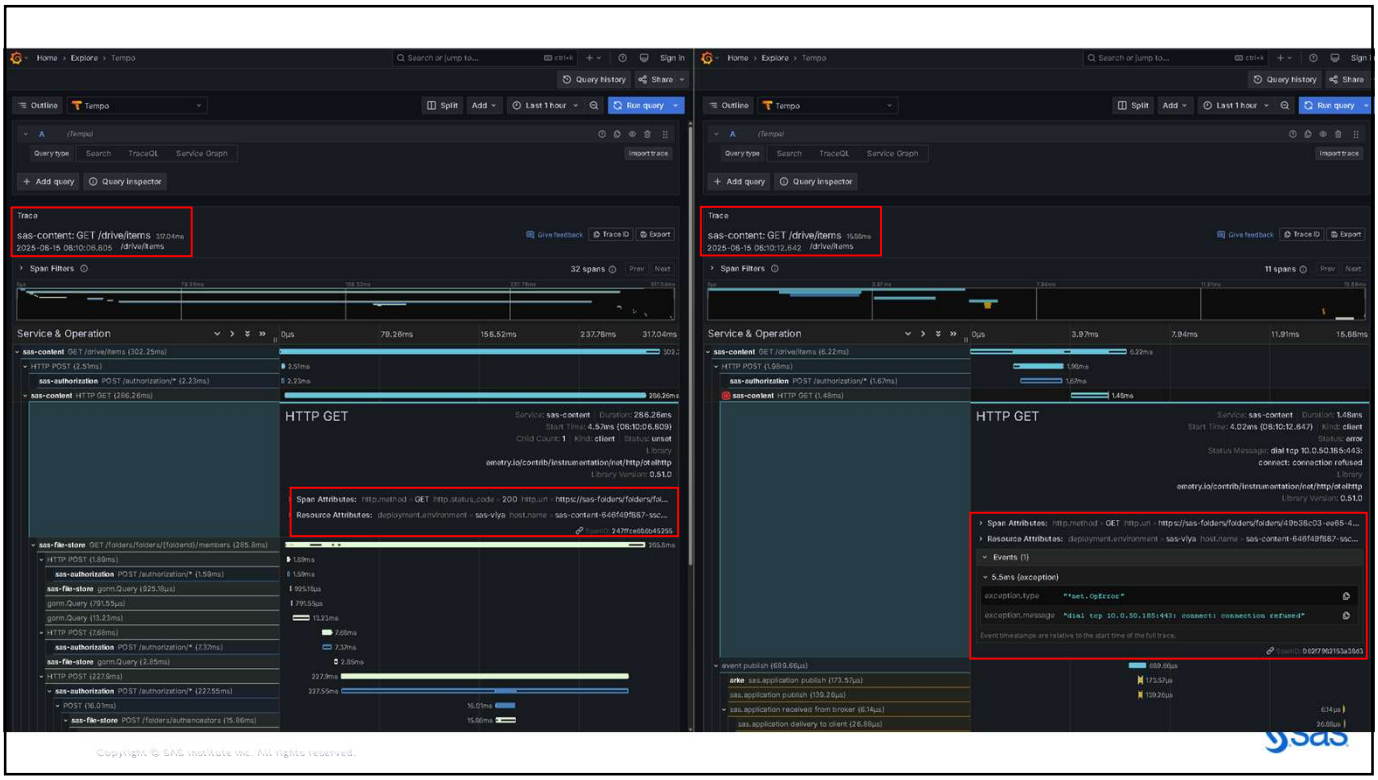

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Solutions

Multi-user, avoiding “resets”

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- Lots of tests had issues and expectations of running alone
  - File names like “test.sas”
  - Zero-state checks
  - User or system settings that change behavior

```
Scenario: Upload files to SAS Server
  Then the user navigates to 'SAS Server/SAS Server/Home/tmp' in the navigation pane
  Then the user clicks the 'Upload files from your local file system' in the nav tree toolbar
  Then the user adds following file(s) to upload dialog: 'CARS.csv,Program_Baseline.sas,StartCAS.sas'
  Then the user waits '5' seconds
  Then the user clicks the 'Upload' button in the 'Upload Files' dialog
  Then the user waits '5' seconds
```

```
Given the 'SASSTUDIO-46343_ContentPaneSearch' feature flag has been temporarily turned on for '360' seconds
Given the 'SASSTUDIO-48268_CalculatePathLabel' feature flag has been temporarily turned on for '360' seconds
```

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Functional vs Performance tests

- Different goals
- Particularly well-written functional tests can cause issues
- Hardcoded 15 second wait times
  - Waits too long – thing is done in 5 seconds!
  - Doesn't wait long enough (or at all) – can add wrappers around it
  - BOTH CASES CAN HAPPEN FOR THE SAME USER ACTION
    - Include screenshot of examples
- Not actually waiting for doneness
- Silver lining – this can help to improve functional tests
  - Wait for UI or specific network traffic instead

```
586 +        if skip_error_check:
            Comment on line R586
587 +            return table_name
588          try:
589              report_alert = Alert(choose_data, "i18n::SAS Visual Analytics")
590              expect(report_alert.error_title).to_be_visible(timeout=5000)
591              logging.debug("That version did not work, trying the next match")
592              if keyboard_only:
593                  tab_to_locator_submit_space(
594                      report_alert, report_alert.locate(button="Close")
595                  )
596              else:
597                  report_alert.locate(button="Close").click()
598          except Exception as _e:
599              # return table name if calling method needs to wait for Data panel
600              return table_name
601
602      # if we get here, no table name matched
603      pytest.fail(f"Could not find suitable match for {table_name}")
604
```

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Costs

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Downsides

- Mixing API + UI tests
- Detect rare errors with deeper understanding of actual-customer-issues
- This isn't free
  - There is effort to be put in in the test container
  - Mixed success
  - Some test containers were more able to pick up the changes

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Finale

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